Mathematical Statistics

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1 Introduction

1.1 Samples and Statistics

Definition 1.1. *(Simple Random Sample)*
A simple random sample is an i.i.d. sample.

Example.
Let $X_1, ..., X_n$ be i.i.d. $N(\mu, \sigma^2)$, and set

$$A = \frac{1}{n}\sum_{i=1}^{n}(X_i - \mu)^2, \quad B = \frac{1}{n-1}\sum_{i=1}^{n}\left(X_i - \overline{X}\right)^2.$$

Then $E(A) = E(B) = \sigma^2$, $\mathrm{Var}(A) = \frac{2}{n}\sigma^4$, $\mathrm{Var}(B) = \frac{2}{n-1}\sigma^4$, and

$$\mathrm{Cor}(A, B) = \sqrt{\frac{n-1}{n}}.$$

Remark.
For an i.i.d. pair,

$$\sigma^2 = E(X - EX)^2 = \frac{1}{2}E(X_1 - X_2)^2 = \iint \frac{1}{2}(x-y)^2 \, \mathrm{d}F(x) \, \mathrm{d}F(y),$$

and $\mu = \int x \, \mathrm{d}F(x)$.

Definition 1.2. *(Parametric Family)*
Let $X_1, ..., X_n$ be i.i.d. with density or mass function f belonging to

$$\mathcal{F} = \{f(x, \theta) : \theta \in \Theta\}.$$

The unknown θ is the parameter, and Θ is the parameter space.

Definition 1.3. *(Statistic)*
A statistic is a quantity determined entirely by the sample, that is, a function of the sample.

Definition 1.4. *(Empirical Distribution Function)*

$$F_n(x) = \frac{1}{n}\sum_{i=1}^{n} I_{\{X_i \leq x\}}.$$

For each fixed x, $E(F_n(x)) = F(x)$ and $\mathrm{Var}(F_n(x)) = \frac{1}{n}F(x)(1-F(x))$.

Theorem 1.1. *(Glivenko-Cantelli)*

$$\sup_x |F_n(x) - F(x)| \to 0$$

almost surely.

Theorem 1.2. *(Dvoretzky-Kiefer-Wolfowitz)*
For every $\varepsilon > 0$,

$$P\left(\sup_x |F_n(x) - F(x)| > \varepsilon\right) \leq 2e^{-2n\varepsilon^2}.$$

Definition 1.5. *(Sample Mean and Sample Variance)*

$$\overline{X} = \frac{1}{n}\sum_{i=1}^{n} X_i, \quad S_n^2 = \frac{1}{n}\sum_{i=1}^{n}\left(X_i - \overline{X}\right)^2.$$

The unbiased sample variance, used below for t and F laws and for confidence intervals, is

$$S^2 = \frac{1}{n-1}\sum_{i=1}^{n}\left(X_i - \overline{X}\right)^2.$$

Theorem 1.3.
If $Z = (Z_1, \ldots, Z_n)^T$ is standard normal in $\mathbb{R}^n$ and U is an orthogonal matrix, then UZ has the same distribution as Z. Consequently, for a normal sample, $\overline{X}$ and S_n^2 are independent.

1.2 Sampling Distributions

Definition 1.6. *(Chi-Square)*
If $X_1, \ldots, X_n$ are i.i.d. $N(0,1)$, then $T = X_1^2 + \ldots + X_n^2$ has the chi-square law χ_n^2 with density

$$f(x) = \frac{1}{2^{\frac{n}{2}}\Gamma\left(\frac{n}{2}\right)} x^{\frac{n}{2}-1} e^{-\frac{x}{2}}, \quad x > 0.$$

$E(T) = n$, $\operatorname{Var}(T) = 2n$, and $E\left(T^k\right) = 2^k \frac{\Gamma\left(\frac{n}{2}+k\right)}{\Gamma\left(\frac{n}{2}\right)}$ whenever $\frac{n}{2} + k > 0$.

Remark.
$\Gamma(t) = \int_0^\infty x^{t-1}e^{-x}\,\mathrm{d}x$, with $\Gamma(1) = 1$, $\Gamma(x+1) = x\Gamma(x)$, $\Gamma\left(\frac{1}{2}\right) = \sqrt{\pi}$, and $\Gamma(n) = (n-1)!$.
If an exponential random variable of rate λ has density $\lambda e^{-\lambda x}$ for $x \geq 0$, then χ_2^2 has the same distribution as the exponential law of rate $\frac{1}{2}$, and also the same distribution as 2λ times an exponential random variable of rate λ.

Definition 1.7. *(Student's t)*
If $X_1, \ldots, X_n, Y$ are i.i.d. $N(0,1)$ and

$$T = \frac{Y}{\sqrt{\frac{X_1^2 + \ldots + X_n^2}{n}}},$$

then T has Student's t law on n degrees of freedom, with density

$$f(t) = \frac{\Gamma\left(\frac{n+1}{2}\right)}{\sqrt{n\pi}\Gamma\left(\frac{n}{2}\right)}\left(1+\frac{t^2}{n}\right)^{-\frac{n+1}{2}}.$$

$E(T) = 0$ for $n > 1$, and $\mathrm{Var}(T) = \frac{n}{n-2}$ for $n > 2$. The case $n = 1$ is standard Cauchy, and as $n \to \infty$ the law tends to $N(0,1)$. The square of a t_n random variable is $F_{1,n}$.

Definition 1.8. *(Fisher's F)*
If $X_1, ..., X_m, Y_1, ..., Y_n$ are i.i.d. $N(0,1)$ and

$$T = \frac{\frac{X_1^2+...+X_m^2}{m}}{\frac{Y_1^2+...+Y_n^2}{n}},$$

then T has the Fisher law $F_{m,n}$, with density

$$f(t) = \frac{\Gamma\left(\frac{m+n}{2}\right)}{\Gamma\left(\frac{m}{2}\right)\Gamma\left(\frac{n}{2}\right)}\left(\frac{m}{n}\right)^{\frac{m}{2}} t^{\frac{m}{2}-1}\left(1+\left(\frac{m}{n}\right)t\right)^{-\frac{m+n}{2}}$$

for $t > 0$. If $n > 2$, then $E(T) = \frac{n}{n-2}$.

Remark.
For normal samples of sizes m and n, with unbiased sample standard deviations S_X and S_Y,

$$\sqrt{m}\frac{\overline{X}-\mu_1}{S_X} \sim t_{m-1}, \quad \sqrt{n}\frac{\overline{Y}-\mu_2}{S_Y} \sim t_{n-1},$$

and $\left(\frac{\sigma_2^2}{\sigma_1^2}\right)\left(\frac{S_X^2}{S_Y^2}\right) \sim F_{m-1,n-1}$.

1.3 Order Statistics

Remark.
If $X_1, ..., X_n$ are i.i.d. with density f, the joint density of the order statistics on $x_1 < ... < x_n$ is $n!f(x_1)...f(x_n)$. If the distribution function F of X is continuous, then $F(X) \sim U(0,1)$.

Proposition 1.1.
Let $U_{(k)}$ be the k-th order statistic of an i.i.d. uniform sample of size n on $(0,1)$. Its density is

$$f(x) = \frac{n!}{(k-1)!(n-k)!} x^{k-1}(1-x)^{n-k}, \quad 0 < x < 1.$$

Hence $E\big(U_{(k)}\big) = \frac{k}{n+1}$ and $\operatorname{Var}\big(U_{(k)}\big) = \frac{k(n+1-k)}{(n+1)^2(n+2)}$. For $1 \le i \le j \le n$,

$$\operatorname{Cov}\big(U_{(i)}, U_{(j)}\big) = \frac{i(n+1-j)}{(n+1)^2(n+2)}, \quad \rho\big(U_{(i)}, U_{(j)}\big) = \sqrt{\frac{i(n+1-j)}{j(n+1-i)}}.$$

Proof.

$$E\big(U_{(k)}\big) = \int_0^1 \frac{n!}{(k-1)!(n-k)!} x^{k}(1-x)^{n-k}\,\mathrm{d}x.$$

The integrand is $\frac{k}{n+1}$ times the density of the $(k+1)$-st order statistic in a uniform sample of size $n+1$, so the integral equals $\frac{k}{n+1}$. The same comparison gives $E\big(U_{(k)}^2\big) = \frac{k(k+1)}{(n+1)(n+2)}$, and the variance follows by subtraction. The covariance is the corresponding mixed moment minus the product of the means. ■

Theorem 1.4.
Fix $p \in (0,1)$ and set $k = \lfloor np \rfloor$. If $U_{(k)}$ is a uniform order statistic, then

$$\sqrt{n}\big(U_{(k)} - p\big) \xrightarrow{d} N(0, p(1-p)).$$

If $X_1, \dots, X_n$ are i.i.d. with continuous distribution function F and density f, and if $f\big(F^{-1}(p)\big) > 0$, then

$$\sqrt{n}\big(X_{(k)} - F^{-1}(p)\big) \xrightarrow{d} N\left(0, \frac{p(1-p)}{\big(f(F^{-1}(p))\big)^2}\right).$$

Lemma 1.1. *(Scheffé)*
Pointwise convergence of densities implies weak convergence of the corresponding distribution functions.

2 Point Estimation

2.1 Estimators

Definition 2.1. *(Unbiased Estimator)*
An estimator $\hat{g}(X)$ of $g(\theta)$ is unbiased if $E_\theta \hat{g}(X) = g(\theta)$ for every θ.

Example.
The unbiased sample variance $S^2 = \frac{1}{n-1}\sum_{i=1}^{n}\left(X_i - \overline{X}\right)^2$ is unbiased for

$$\sigma^2 = \frac{1}{2}\iint (x-y)^2 \,\mathrm{d}F(x)\,\mathrm{d}F(y).$$

Definition 2.2. *(Mean Squared Error)*

$$\mathrm{MSE}\left(\hat{\theta}\right) = E\left(\hat{\theta}-\theta\right)^2 = \mathrm{Var}\left(\hat{\theta}\right) + \left(E\hat{\theta}-\theta\right)^2.$$

The second term is the square of the bias.

Definition 2.3. *(Efficiency)*
Let $\hat{g}_1$ and $\hat{g}_2$ be unbiased estimators of $g(\theta)$. If $\mathrm{Var}(\hat{g}_1(X)) \le \mathrm{Var}(\hat{g}_2(X))$ for every θ, and the inequality is strict for some θ_0, then $\hat{g}_1$ is more efficient than $\hat{g}_2$.

Definition 2.4. *(Consistency)*
An estimator $\hat{g}_n$ is weakly consistent for $g(\theta)$ if $\hat{g}_n \xrightarrow{P} g(\theta)$, strongly consistent if the convergence is almost sure, and completely convergent if

$$\sum_{n} P(|\hat{g}_n - g(\theta)| > \varepsilon) < \infty$$

for every $\varepsilon > 0$.

Remark.
The weak law, Markov's inequality, and Chebyshev's inequality are the usual tools for weak consistency. The strong law and the Borel–Cantelli lemma are the usual tools for strong consistency.

Theorem 2.1. *(Continuous Mapping)*
If g is continuous and X_n converges to X almost surely or in probability, then

$g(X_n)$ converges to $g(X)$ in the same mode. If $X_n \xrightarrow{d} X$ and $P(X \in C) = 1$ for the continuity set C of g, then $g(X_n) \xrightarrow{d} g(X)$.

Definition 2.5. *(Maximum Likelihood)*
The likelihood of an i.i.d. sample is $L(\theta, x) = \prod_{i=1}^{n} f(x_i, \theta)$. A maximum likelihood estimator is a value of θ that maximizes L.

Remark.
Two parallel procedures, starting from a model:

- Method of moments. Write $E(X) = f(p)$, invert to $p = g(E(X))$, and substitute the sample mean to get $\hat{p} = g\left(\overline{X}\right)$.
- Maximum likelihood. Write $L(p)$, solve $L'(p) = 0$, and check that the critical point is a maximum through the Hessian.

Definition 2.6. *(UMVUE)*
An unbiased estimator $\hat{g}^*$ of $g(\theta)$ is a uniformly minimum variance unbiased estimator if $\mathrm{Var}(\hat{g}^*(X)) \leq \mathrm{Var}(\hat{g}(X))$ for every unbiased competitor $\hat{g}$ and every θ.

Example.
If $X \sim \mathrm{Bin}(n, p)$ and $g(p) = \frac{1}{p}$, then g has no unbiased estimator. Indeed, an identity

$$\sum_{k=0}^{n} \hat{g}(k) \binom{n}{k} p^{k+1} (1-p)^{n-k} = 1$$

for every $p \in (0, 1)$ is impossible: as $p \to 0$ the left side tends to 0.

2.2 Sufficiency and Completeness

Theorem 2.2. *(Rao-Blackwell)*
Let $\hat{g}(X)$ be unbiased for $g(\theta)$, let $T = T(X)$, and set $h(T) = E(\hat{g}(X) \mid T)$. Then $Eh(T) = g(\theta)$ and $\mathrm{Var}(h(T)) \leq \mathrm{Var}(\hat{g}(X))$ for every θ, with equality if and only if $\hat{g}(X) = h(T)$ almost surely. If the conditional law of the sample given T does not depend on θ, then T is sufficient for θ.

Remark.
If $f(X) = E(Y|X)$, then $\mathrm{Var}(Y) = \mathrm{Var}(f(X)) + E(Y - f(X))^2$.

Theorem 2.3. *(Factorization)*
A statistic T is sufficient for θ if and only if the joint density or mass function factors as $f(x, \theta) = g(T(x), \theta)h(x)$.

Definition 2.7. *(Complete Statistic)*
A statistic T is complete if $E_\theta \varphi(T) = 0$ for every θ implies $P_\theta(\varphi(T) = 0) = 1$ for every θ.

Remark.
A standard problem is: given the distribution, and possibly a sufficient statistic T, find the UMVUE of $g(\theta)$.

Theorem 2.4.
An unbiased estimator $\hat{g}$ is a UMVUE of $g(\theta)$ if and only if $\mathrm{Cov}(\hat{g}(X), Z(X)) = E(\hat{g}(X)Z(X)) = 0$ whenever $E_\theta Z(X) = 0$ for every θ.

Theorem 2.5. *(Lehmann--Scheffé)*
Suppose $g(\theta)$ admits an unbiased estimator and T is a complete sufficient statistic. If $\hat{g}(T)$ is unbiased for $g(\theta)$, then $\hat{g}(T)$ is the unique UMVUE of $g(\theta)$.

Remark.
Two ways to produce a UMVUE: apply the Lehmann–Scheffé theorem to an unbiased function of a complete sufficient statistic, often computed from $E(T)$ and $E(T^2)$; or start from any unbiased estimator and replace it by its conditional expectation given a sufficient statistic.

2.3 The Cramér-Rao Bound

Definition 2.8. *(Fisher Information)*
For a density or mass function $f(x, \theta)$,

$$\int \frac{\left(\frac{\partial f(x,\theta)}{\partial \theta}\right)^2}{f(x,\theta)}\,\mathrm{d}x = E\left(\frac{\partial \log f(X,\theta)}{\partial \theta}\right)^2 =: I(\theta).$$

This is the Fisher information in one observation.

Theorem 2.6. *(Cramér-Rao)*
Let the model be Cramér-Rao regular, and let $\hat{g}(X)$ be unbiased for $g(\theta)$. If $\int \hat{g}(x) f(x,\theta)\,\mathrm{d}x$ may be differentiated in θ under the integral, then

$$\operatorname{Var}(\hat{g}(X)) \geq \frac{\left(g'(\theta)\right)^2}{nI(\theta)}.$$

Remark.
The usual route to a UMVUE through this bound is: compute $I(\theta)$, apply the inequality, and check whether an unbiased function of a sufficient statistic attains the bound.

Definition 2.9. *(Entropy and Kullback-Leibler Divergence)*
The entropy of a density $f(x,\theta)$ is $E(-\log f(X,\theta))$. The divergence of p from q is

$$D_{\mathrm{KL}}(p \parallel q) = \int \log\left(\frac{p(x)}{q(x)}\right) p(x)\,\mathrm{d}x.$$

Remark.
Under the usual regularity, a maximum likelihood estimator $\hat{\theta}_n$ converges to the population maximizer $\theta^* = \operatorname{argmax}_\theta E \log f(X,\theta)$.

3 Interval Estimation

Definition 3.1. *(Confidence Interval)*
Random endpoints $\hat{\theta}_1 < \hat{\theta}_2$ form a confidence interval for θ of level $1-\alpha$ if $P\left(\theta \in \left(\hat{\theta}_1, \hat{\theta}_2\right)\right) \geq 1-\alpha$. One often takes $\alpha = 5\%$.

Remark.
Let $X_1, \ldots, X_n$ be i.i.d. $N(\mu, \sigma^2)$. Write $u(\alpha) = \Phi^{-1}(1-\alpha)$ for the upper α quantile of $N(0,1)$, and write $t_{n-1}(\alpha)$ and $\chi^2_\nu(\alpha)$ for the corresponding upper quantiles of t_{n-1} and χ^2_ν. Let S be the unbiased sample standard deviation.

- If σ is known, a level $1-\alpha$ interval for μ is

$$\left[\overline{X} - \left(\frac{\sigma}{\sqrt{n}}\right) u\left(\frac{\alpha}{2}\right), \overline{X} + \left(\frac{\sigma}{\sqrt{n}}\right) u\left(\frac{\alpha}{2}\right)\right].$$

- If σ is unknown, replace σ by S and $u(\frac{\alpha}{2})$ by $t_{n-1}(\frac{\alpha}{2})$.
- If μ is known, a level $1-\alpha$ interval for σ^2 is

$$\left[\frac{\sum_{i=1}^{n}(X_i-\mu)^2}{\chi_n^2}\left(\frac{\alpha}{2}\right), \frac{\sum_{i=1}^{n}(X_i-\mu)^2}{\chi_n^2}\left(1-\frac{\alpha}{2}\right)\right].$$

- If μ is unknown, replace each $(X_i - \mu)$ by $\left(X_i - \overline{X}\right)$ and each χ_n^2 by χ_{n-1}^2.

Definition 3.2. *(Pivotal Quantity)*
Starting from a point estimator $T = \hat{\theta}$, form a function $\psi(T, \theta)$ whose distribution does not depend on the unknown parameter. Choose constants $a < b$ with $P(a \leq \psi(T,\theta) \leq b) = 1 - \alpha$, and invert the inequality for θ.

4 Hypothesis Testing

4.1 Tests and Likelihood Ratios

Definition 4.1. *(Test, Power, and Errors)*
Let $X_1, ..., X_n$ have joint law $f(x, \theta)$. A test of $H_0 : \theta \in \Theta_0$ against $H_1 : \theta \in \Theta_1$ rejects H_0 when the sample falls in a rejection region A. The power function is $\beta(\theta) = P_\theta(X \in A)$. Rejecting a true null hypothesis is a type I error. Accepting a false null hypothesis is a type II error. The test has level α if $\sup_{\theta\in\Theta_0} \beta(\theta) \leq \alpha$.

Definition 4.2. *(Test Function)*
The test function is $\varphi(X) = I_{\{X\in A\}}$. For a test statistic T and a critical value c, one takes $A = \{T(X) > c\}$.

Definition 4.3. *(Likelihood-Ratio Statistic)*

$$T_n = \frac{\sup_{\theta\in\Theta} L(\theta)}{\sup_{\theta\in\Theta_0} L(\theta)}.$$

One has $T_n \geq 1$, and values closer to 1 favor the null. The test rejects for large values. A possibly randomized test function is

$$\varphi(x) = \begin{cases} 1 \text{ if } T_n > c \\ r \text{ if } T_n = c, \\ 0 \text{ if } T_n < c \end{cases}$$

where $0 < r < 1$ and c is chosen to meet the level.

Remark.
As $n \to \infty$, $2 \log T_n$ converges in distribution to a chi-square law.

Definition 4.4. *(p-Value)*
Under the null hypothesis, if G is the distribution function of the test statistic, then $G(T) \sim U(0,1)$. The p-value is $1 - G(T)$. In the notes, G may be the asymptotic null distribution.

4.2 Rank Tests

Definition 4.5. *(Rank Statistic)*
If $X_1, ..., X_n$ are distinct, the rank R_k of X_k is its position in the increasing rearrangement, and $R = (R_1, ..., R_n)$ is the rank statistic. Then $R_i = \sum_{k=1}^{n} I_{\{X_k \leq X_i\}} = nF_n(X_i)$.

Theorem 4.1.
If $X_1, ..., X_n$ are i.i.d. with a continuous distribution, then R is uniformly distributed on the permutations of $(1, ..., n)$: each permutation has probability $\frac{1}{n!}$. Marginally, $P(R_1 = k) = \frac{1}{n}$,

$$E(R_i) = \frac{n+1}{2},$$

$$\text{Var}(R_i) = \frac{(n+1)(n-1)}{12},$$

$$\text{Cov}(R_i, R_j) = -\frac{n+1}{12}$$

for $i \neq j$.

Definition 4.6. *(Signed Rank Statistic)*
Let $\varphi_i = \text{sgn}(X_i)$, and let R_i be the rank of $|X_i|$ among $|X_1|, ..., |X_n|$. The signed rank statistic is $R^+ = (\varphi_1 R_1, ..., \varphi_n R_n)$.

Theorem 4.2. *(Chi-Square Goodness of Fit)*
For a fully specified discrete distribution with cell probabilities $p_i > 0$, if $\hat{p}_i$ is the observed frequency of cell i in a sample of size n, then as $n \to \infty$,

$$\sum_{i=1}^{k} \frac{n(\hat{p}_i - p_i)^2}{p_i} \xrightarrow{d} \chi^2_{k-1}.$$

Theorem 4.3. *(Kolmogorov-Smirnov)*
If F is a continuous distribution function and F_n is the empirical distribution function of an i.i.d. sample from F, then

$$\sqrt{n} \sup_x |F_n(x) - F(x)| \xrightarrow{d} \sup_t |B(t)|,$$

where B is a Brownian bridge.